

Technical Architecture Brief: Macro-Hedging SPV Simulation Engine

Project Objective: To develop a stochastic simulation and scenario-testing engine that models the financial lifecycle of the Synthetic Sovereign Hedge (UK FIC vs. Kenyan SPV). The simulator will validate the Debt Service Coverage Ratio (DSCR) for the UK FIC, plot the Pareto-optimal frontier for SPV investors across Class A, B, and C tranches, and stress-test the model against extreme macroeconomic volatility via Monte Carlo simulations.

1. System Architecture & Tech Stack Recommendations

Given the requirement for mathematical precision, matrix operations, and event-driven state changes, the following stack is recommended:

- **Core Engine (Backend):** Python 3.10+
 - *Libraries:* NumPy & SciPy (for stochastic differential equations and Monte Carlo paths), Pandas (for time-series cash flow ledgers).
- **API Layer:** FastAPI (to serve simulation results to a frontend dashboard).
- **Frontend (Optional Dashboard):** React (Next.js) with Recharts or Plotly.js to visualize the Pareto frontier, FX degradation curves, and FIC cash flow.

2. Core Simulation Modules

The engine must be divided into three interacting modules that step through time at discrete intervals (Monthly Ticks, t_0 to t_{120} for a 10-year max horizon).

Module A: The Macroeconomic Environment (The Global State)

This module generates the exogenous variables that drive the simulation.

- **FX Generator (KES/GBP):** Implement a Jump-Diffusion model (e.g., Merton's model) rather than a standard Geometric Brownian Motion. A Jump-Diffusion model accounts for standard daily volatility *plus* sudden, massive devaluation events (the "jumps" mimicking sovereign default).
- **Domestic Bond Yield Tracker:** Simulates the alternative risk-free rate in Kenya (e.g., hovering at 16% - 20%).
- **UK UK Property Market:** Simulates UK house price index (HPI) growth and rental inflation for Woolston/Southampton.

Module B: The UK FIC Ledger (Borrower State)

This module tracks the operational cash flow, tax liabilities, and real estate equity of the UK entity.

- **Inputs:** Property Value (£360,000), Gross Rent (£1,800/mo), Management Fees (14.4%), Base KES Interest Rate (3%).
- **Tax Engine:** Calculates UK Corporation Tax on net rental profits (allowing deductions for the 3% interest and engineered early repayment fees) and computes Withholding Tax on outbound SPV payments.
- **Refinance Trigger:** An event listener that activates when a debt tranche is called. Calculates available equity and simulates a UK BTL mortgage approval at prevailing UK interest rates (e.g., 5.5% - 6.5%).

Module C: The SPV Tranche State Machine (Lender State)

This module manages the underlying 70,000,000 KES debt and tracks the exact status of the 14 Investor Packs.

- **Tranche Listener:** Constantly monitors the Macro Environment FX rate.
 - *If KES/GBP drops 20%:* Change State of Class A Packs to `CALLED` .
 - *If KES/GBP drops 35%:* Change State of Class B Packs to `CALLED` .
 - *If KES/GBP drops 50%:* Change State of Class C Packs to `CALLED` .
- **Payout Calculator:** Upon a `CALLED` state, executes the payout function:
$$Y_{Total} = I_{Base} + F_{ER} + (\Pi_{FX} \times \%_{Share}).$$
- **Revenue Share Tracker:** If Class C is called, initializes a 24-month countdown timer, routing a percentage of FIC rental income back to the SPV.

3. Configurable Simulation Parameters (The UI Controls)

The simulator must expose the following variables for dynamic scenario testing:

Macro Variables:

- `kes_gbp_initial_spot` : Default 166.6
- `devaluation_jump_probability` : % chance of crash occurring in any given month.
- `jump_severity_mean` : Default -30%.

FIC / Property Variables:

- `property_initial_value` : Default £360,000
- `monthly_rent_gbp` : Default £1,800
- `uk_btl_mortgage_rate` : Default 6.0%

Tranche Variables:

- `spv_total_raise_kes` : Default 70,000,000
- `class_a_trigger` / `class_a_sweetener`
- `class_b_trigger` / `class_b_sweetener`
- `class_c_trigger` / `class_c_sweetener`

4. Output Metrics & Visualizations

To prove the Pareto Optimal Equilibrium to institutional investors, the simulation engine must output the following specific charts and datasets:

1. **The Investor IRR vs. Domestic Bond Chart:** A scatter plot comparing the simulated Internal Rate of Return (IRR) of Class B and C packs (in GBP) against the simulated domestic Kenyan sovereign bond yields across 10,000 scenarios.
2. **FIC DSCR Timeline:** A time-series graph proving the UK FIC maintains a Debt Service Coverage Ratio > 1.25 across the holding period, ensuring the property never operates at a loss.
3. **FIC Net Wealth Accumulation:** A stacked area chart showing the transition of the FIC's liabilities from SPV KES Debt -> UK BTL GBP Debt, and the corresponding explosion in the FIC's retained corporate equity upon tranche execution.
4. **Capital Flight Survival Rate:** A counterfactual metric showing what 70M KES would be worth in GBP if left in Kenya (subject to inflation and devaluation) vs. its final settlement value via the

SPV.

5. Development Roadmap (Next Steps for Quant Team)

- **Phase 1 (Logic Sandbox):** Build a deterministic Python script to map out one perfectly executed scenario (e.g., a smooth 35% crash in Month 24) to validate the core accounting formulas.
- **Phase 2 (Stochastic Implementation):** Introduce the Jump-Diffusion FX model and run a 1,000-path Monte Carlo simulation to find edge cases (e.g., what happens if the currency drops 15% and stays flat for 10 years?).
- **Phase 3 (Tax & Legal Nuance):** Integrate UK Corporation Tax ledgers and Double Taxation Treaty (DTA) withholding tax offsets to ensure net cash flow projections are perfectly compliant.